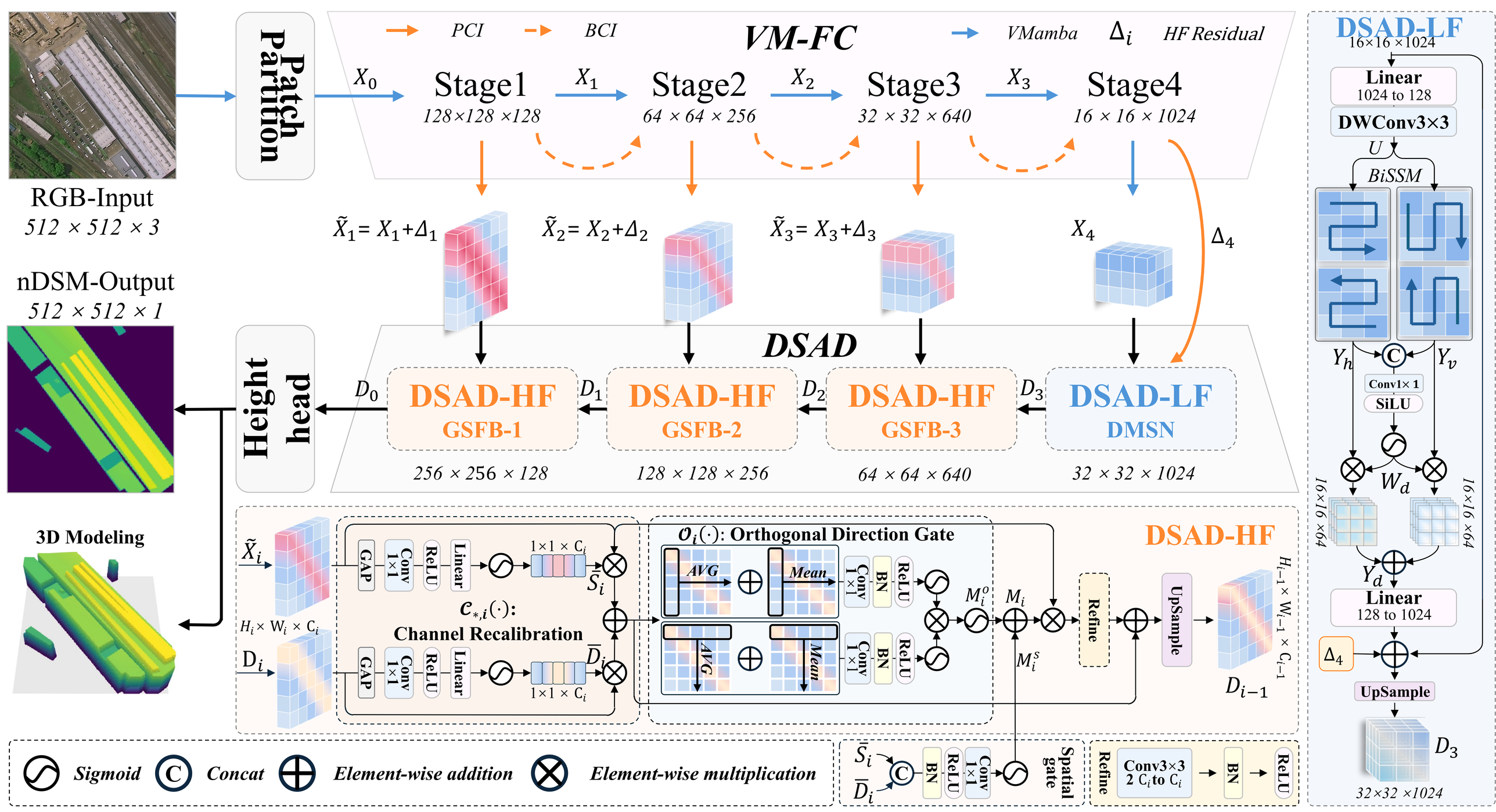

RGB-Input
512 × 512 × 3
nDSM-Output
512 × 512 × 1
3D Modeling
Patch Partition
PCI
BCI
VM-FC
VMamba
Δ_i
HF Residual
X_0
Stage1
128×128 ×128
X_1
Stage2
64 × 64 × 256
X_2
Stage3
32 × 32 × 640
X_3
Stage4
16 × 16 × 1024
X_4
Δ_4
$\tilde{X}_1 = X_1 + \Delta_1$
$\tilde{X}_2 = X_2 + \Delta_2$
$\tilde{X}_3 = X_3 + \Delta_3$
DSAD
Height head
D_0
D_1
D_2
D_3
DSAD-HF
GSFB-1
DSAD-HF
GSFB-2
DSAD-HF
GSFB-3
DSAD-LF
DMSN
256 × 256 × 128
128 × 128 × 256
64 × 64 × 640
32 × 32 × 1024
DSAD-LF
16×16 ×1024
Linear
1024 to 128
DWConv3×3
U
BiSSM
Y_h
C
Y_v
Conv1× 1
SiLU
W_d
16×16 ×64
16×16 ×64
Y_d
Linear
128 to 1024
Δ_4
UpSample
D_3
32×32 ×1024
DSAD-HF
$\tilde{X}_i$
$H_i \times W_i \times C_i$
D_i
GAP
Conv 1×1
ReLU
Linear
1×1 × C_i
$\bar{S}_i$
$\mathcal{C}_{*,i}(\cdot)$:
Channel Recalibration
$\bar{D}_i$
$\mathcal{O}_i(\cdot)$: Orthogonal Direction Gate
AVG
Mean
Conv 1×1
BN
ReLU
M_i^o
M_i
M_i^s
Refine
UpSample
$H_{i-1} \times W_{i-1} \times C_{i-1}$
D_{i-1}
Sigmoid
Concat
Element-wise addition
Element-wise multiplication
Spatial gate
Conv3×3
2 C_i to C_i